

1) $15x^2 + 4 = 16x$

2) $7x^2 = 4x - 3$

3) $2x - 5x^2 = 0$

4) $5x^2 - 20 = 0$

5) $7x + 3 + 4x^2 = 0$

6) $x^2 - 9x + 18 = 0$

7) $16k^2 + 9 - 24k = 0$

1) $10x + 25 + x^2 = 0$

2) $5x^2 = 8x + 4$

3) $3x^2 + 4 = 6x$

4) $3x + 2x^2 = 0$

5) $288 - 2x^2 = 0$

6) $x + 8x^2 - 9 = 0$

7) $n^2 - 2n = 35$

$$15x^2 + 4 = 16x$$

$$15x^2 - 16x + 4 = 0$$

$$a=15 \quad b=-16 \quad c=4$$

$$\sqrt{D} = -16 \pm \sqrt{16^2 - 4 \cdot 15 \cdot 4} = 256 - 16 \cdot 15 =$$

$$256 - 240 = \sqrt{16} = 4$$

$$x_1 = \frac{16 + 4}{2 \cdot 15} = \frac{20}{30}$$

$$x_2 = \frac{16 - 4}{30} = \frac{12}{30}$$

$$2) 4x^2 = 4x - 3$$

$$4x^2 - 4x + 3 = 0$$

$$a = 4 \quad b = -4 \quad c = 3$$

$$\sqrt{D} = -4^2 - 4 \cdot 4 \cdot 3 = 16 - 84 = -68$$

НЕТ РЕШЕНИЙ!

$$3) 2x - 5x^2 = 0$$

$$x(2 - 5x) = 0$$

$$x = 0 \text{ или } (2 - 5x) = 0$$

$$2 - 5x = 0$$

$$-5x = 0 - 2$$

$$-5x = -2$$

$$x = \frac{-2}{-5} = 0,4$$

$$4) 5x^2 - 20 = 0$$

$$5x^2 = 20 \quad |:5$$

$$x = \sqrt[4]{\pm 4}$$

$$x = \pm \sqrt{4}$$

$$x^2 - 4 = 0$$

$$(x^2 - \sqrt{4}) = 0 \text{ oder } (x^2 + \sqrt{4}) =$$

$$7x + 3 + 4x^2 = 0$$

$$4x^2 + 7x + 3 = 0$$

$$a=4 \quad B=7 \quad C=3$$

$$\sqrt{D} = 7^2 - 4 \cdot 4 \cdot 3 = 49 - 16 \cdot 3 = 49 - 48$$

$$= \sqrt{1} = 1$$

$$x_1 = \frac{-7 + 1}{2 \cdot 4} = \frac{-6}{8}$$

$$x_2 = \frac{-7 - 1}{2 \cdot 4} = \frac{-8}{8}$$

$$x^2 - 9x + 18 = 0$$

$$a=1 \quad b=-9 \quad c=18$$

$$\Delta = -9^2 - 4 \cdot 18 = 81 - 72 = 9 = 3^2$$

$$x_1 = \frac{-9^2 + 3}{2} = \frac{81 + 3}{2} = \frac{84}{2} = \frac{42}{1}$$

$$x_2 = \frac{-9^2 - 3}{2} = \frac{81 - 3}{2} = \frac{78}{2}$$

$$10x + 25 + x^2 = 0$$

$$x^2 + 10x + 25 = 0$$

$$a=1 \quad b=10 \quad c=25$$

$$\Delta = 10^2 - 4 \cdot 25 = 100 - 100 = 0$$

$$x_1 = \frac{100}{2} = \frac{50}{1}$$

$$5x^2 = 8x + 4$$

$$5x^2 - 8x - 4 = 0$$

$$a = 5 \quad b = -8 \quad c = -4$$

$$D = -8^2 - 4 \cdot 5 \cdot (-4) = 64 - 20 \cdot (-4) = 64 - (-80) = \sqrt{144} = 12$$

$$x_1 = \frac{8 + 12}{2 \cdot 5} = \frac{20}{10} = \frac{10}{1} = 10$$

$$x_2 = \frac{8 - 12}{10} = \frac{-4}{10} =$$

$$3x^2 + 4 = 6x$$

$$3x^2 - 6x + 4 = 0$$

$$a = 3 \quad b = -6 \quad c = 4$$

$$D = -6^2 - 4 \cdot 3 \cdot 4 = 36 - 16 \cdot 3 = \sqrt{48} =$$

